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An approach to unsymmetrical 3,3′-diindolylmethanes through Pd-catalyzed cascade Heck cyclization of allenamides and *o*-ethynylanilines†

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A highly efficient synthesis of unsymmetrical 3,3′-diindolylmethanes bearing various substituents has been developed by palladium-catalyzed cascade Heck/cyclization reaction. In this transformation, precursors allenamides and *o*-ethynylanilines were utilized to generate *in situ* indole skeletons. Modification of the methylene unit of diindolylmethane was achieved by treatment with oxidant and silyl nucleophiles. The methodology allows formation of one C–N bond, two C–C bonds and two new rings in one-pot.

Introduction

Diindolylmethanes (DIMs) are one class of unique and privileged indole alkaloids among numerous natural products, exhibiting a wide range of biological activities,[1] such as anti-inflammatory,[1a,b] antiproliferative,[1c] antihyperlipidemic,[1d] anti-cancer[1e] and GPR84 agonistic activity.[1f] Moreover, DIMs are also recognized as core building blocks in the research of colorimetric chemosensors to prepare conjugated tris(indolyl) methane anion receptors[2] and oxidized-bisindole chromophores[3] (Fig. 1).

Owing to their exceptionally potent applications in materials and pharmaceuticals, the preparation of DIMs has attracted intensive attention, and several highly efficient methods have been developed, in particular those that yield symmetric DIMs.[4–14] Brønsted or Lewis acids catalysis is the most straightforward one, wherein key skeleton of DIMs is forged *via* the reaction of indoles and alkyl sources, encompassing carbonyl compounds,[5] amino acids,[6] alkenes,[7] allenes[8] and alkynes[9] (Scheme 1a). Besides, photocatalysis,[10] transition metal catalysis,[11] enzyme catalysis,[12] as well as electrochemical technique[13] have been demonstrated to be effective strategies for construction of symmetric diindolylmethane skeletons (Scheme 1a). In recent years, various research groups have reported the synthesis of unsymmetrical DIMs using a leaving group strategy,[14] which is proposed to proceed through *in situ* generation of stabilized alkylideneindoleninium ion intermediates derived from 3-substituted indoles (Scheme 1b). Nevertheless, the majority of existing route employed elaborately pre-functionalized indoles as the reactants to assemble the symmetrical DIMs system. Thus, exploring an effective and practical strategy to rapidly access diversely unsymmetrical DIMs is still highly desirable, yet challenging. Based on our continuous interests in the exploration of indole containing polycyclic synthesis,[15] herein, we will disclose an unprecedented palladium-catalyzed cascade Heck/cyclization reaction between allenamides[16] and *o*-ethynylanilines[17] to afford unsymmetrically substituted 3,3′-diindolylmethane frameworks. One C–N bond and two C–C bonds formation can be achieved within one-pot procedure (Scheme 1c).

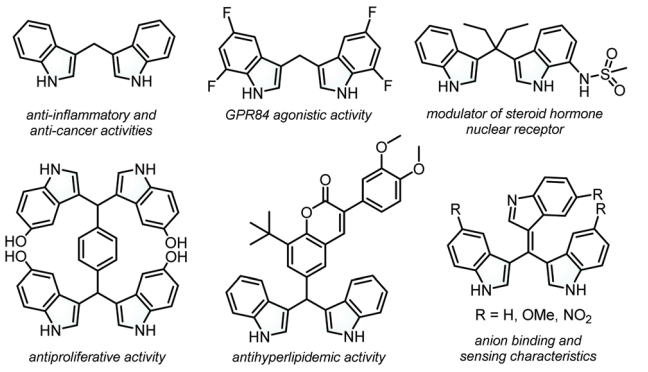


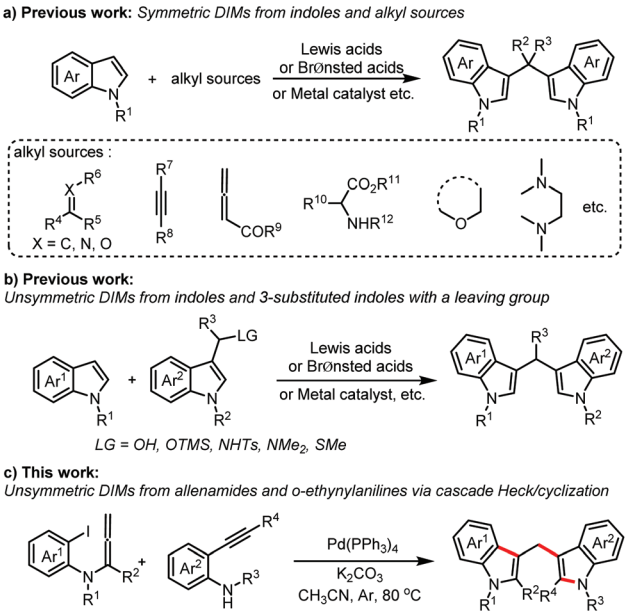
Fig. 1 Representative examples of bioactive compounds and chromophores containing 3,3′-DIMs.

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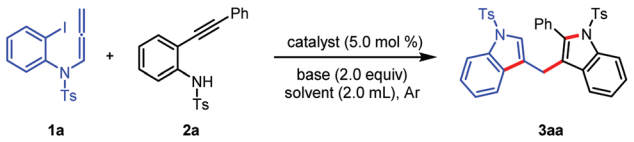
Scheme 1 Strategies to access 3,3′-diindolylmethanes.

Results and discussion

Initial studies were performed with the benchmark substrates *N*-allenyl-2-iodoaniline **1a** and 4-methyl-*N*-(2-(phenylethynyl) phenyl)benzenesulfonamide **2a**. Various palladium catalysts were first tested (Table 1, entries 1–7). When the reaction was conducted with $Pd(PPh_3)_4$ as the catalyst in the presence of Na_2CO_3 in DMF, the desired product **3aa** was isolated in 83% yield (Table 1, entry 4). The investigation of bases revealed that K_2CO_3 was an ideal choice, which afforded **3aa** in 88% yield (Table 1, entries 8–13). The results would be most satisfying wherein acetonitrile was applied as solvent, while others (MeOH, DCE, toluene, THF) led to deterioration of the yield (Table 1, entries 11 and 14–18). In addition, lowering or raising the reaction temperature resulted in poor conversion (Table 1, entries 19 and 20). No expected product could be detected in the absence of palladium catalyst (Table 1, entry 21).

The scope of this cascade Heck/cyclization reaction was next probed, and the results are summarized in Table 2. Regarding to the *N*-aryl-*N*-allenamide **1**, we found that different *N*-protected groups were compatible, the corresponding products (**3aa–3da**) could be isolated in moderate to good yields. Various substituents, such as methoxyl, methyl, halogens (F, Cl, Br), and cyano, were well-tolerated to deliver the products (**3ea–3na**) in 70–98% yields. The methyl-substituted allene **1o** could be converted to **3oa** in 78% yield. Notably, to further increase the molecular diversity, the substrate with pyridine was also investigated, giving **3pa** in 92% yield. A series of *o*-alkynylanilines bearing substituents with different electronic properties at R^4 participated the reaction smoothly to deliver the DIMs (**3ab–3aj**) in 57–98% yields. The

Table 1 Optimization of the reaction conditions[a]



1a 2a 3aa

Entry	Catalyst	Base	Solvent	Yield[b] (%)
1	$Pd_2(dba)_3$	Na_2CO_3	DMF	18
2	$Pd_2(dba)_3{\cdot}CHCl_3$	Na_2CO_3	DMF	30
3	$Pd_2(dppp)_2$	Na_2CO_3	DMF	NR[c]
4	$Pd(PPh_3)_4$	Na_2CO_3	DMF	83
5	$Pd(CH_3CN)_2Cl_2$	Na_2CO_3	DMF	23
6	$Pd(PPh_3)_2Cl_2$	Na_2CO_3	DMF	78
7	$Pd(OAc)_2$	Na_2CO_3	DMF	43
8	$Pd(PPh_3)_4$	K_3PO_4	DMF	81
9	$Pd(PPh_3)_4$	$KHCO_3$	DMF	76
10	$Pd(PPh_3)_4$	Li_2CO_3	DMF	37
11	$Pd(PPh_3)_4$	K_2CO_3	DMF	88
12	$Pd(PPh_3)_4$	Cs_2CO_3	DMF	80
13	$Pd(PPh_3)_4$	NaOAc	DMF	82
14	$Pd(PPh_3)_4$	K_2CO_3	MeOH	42
15	$Pd(PPh_3)_4$	K_2CO_3	DCE	86
16	$Pd(PPh_3)_4$	K_2CO_3	Toluene	37
17	$Pd(PPh_3)_4$	K_2CO_3	CH_3CN	97
18	$Pd(PPh_3)_4$	K_2CO_3	THF	28
19	$Pd(PPh_3)_4$	K_2CO_3	CH_3CN	81[d]
20	$Pd(PPh_3)_4$	K_2CO_3	CH_3CN	75[e]
21	—	K_2CO_3	CH_3CN	NR

[a] Reaction conditions: **1a** (0.1 mmol), 2**a** (0.11 mmol), catalyst (5.0 mol%), base (2.0 equiv.), solvent (2.0 mL), at 80 °C under Ar atmosphere for 24 h, sealed tube. [b] Isolated Yields. [c] NR = no reaction. [d] 60 °C. [e] 100 °C.

structure of the product **3ab** was identified by X-ray crystallography (see the ESI† for details). It is noteworthy that silyl (**3al**) and thiophene (**3am**) groups were well tolerated. Products **3an–3as** with methyl, methoxyl, trifluoromethyl and halogen groups (–F, –Br) groups at the C4- or C5-position of the arylamine were achieved in 50–91% yields. The *o*-alkynylaniline **2t** with methyl at the C6-position could not offer the desired product **3at**. When the substrate **2u** with CF_3CO group was examined, the nitrogen atom deprotected product **3au** was obtained in 50% yield.

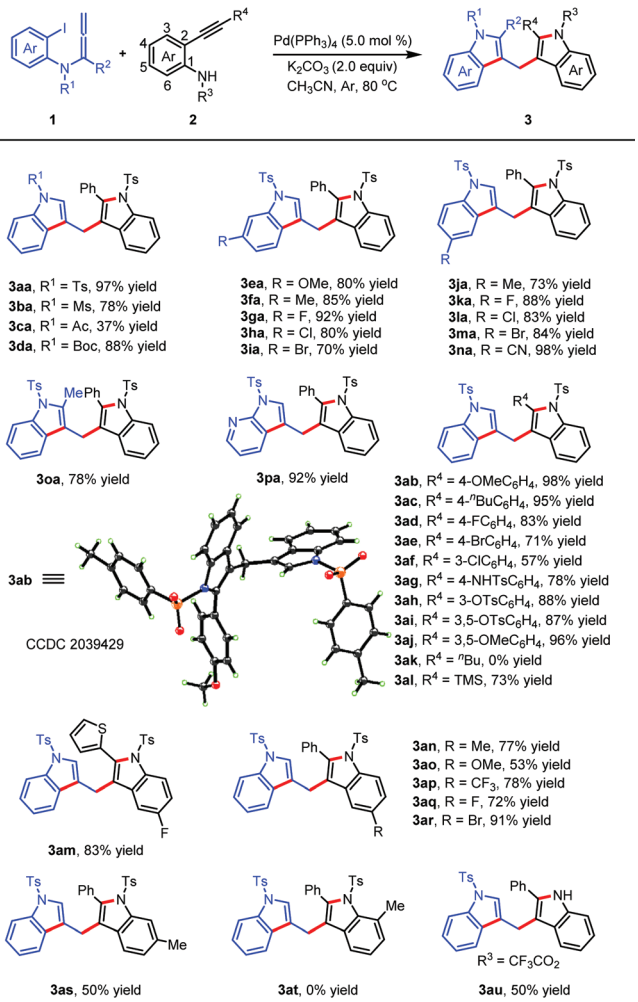
In addition to indole scaffolds, other intriguing heteroarenes, including isoquinolone **5** and benzofuran **7** were all formed in 90% and 43% yield respectively by altering the allene chains (Scheme 2).

Subsequently, to explore the potential utility of this reaction, the scale-up experiments and further transformations of the diindolylmethane products were conducted as shown in Table 3. The compounds **3aa** (1.14 g) and **3al** (0.45 g) could be prepared in 90% and 72% yield respectively on a millimole scale (Table 3a). The TMS and sulfamide groups in **3al** could be removed in the present of NaOH, leading to the symmetric diindolylmethane **8** in 77% yield (Table 3b).[15a] Diindolylmethane **3aa** could be smoothly oxidized to **9** with good efficiency (Table 3c).[18] Compound **10** with a hydroxyl group was readily achieved in 85% yield through the reduction of **9** (Table 3d),[19] which served as a versatile intermediate for

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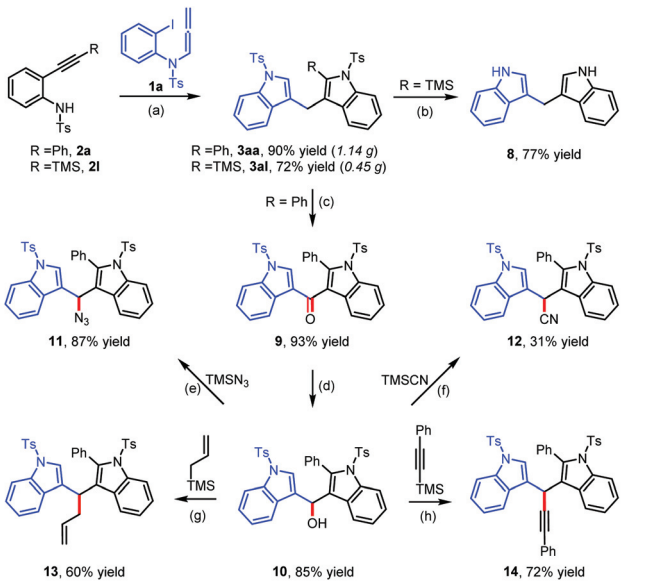
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Table 2 Substrate scope[a,b]



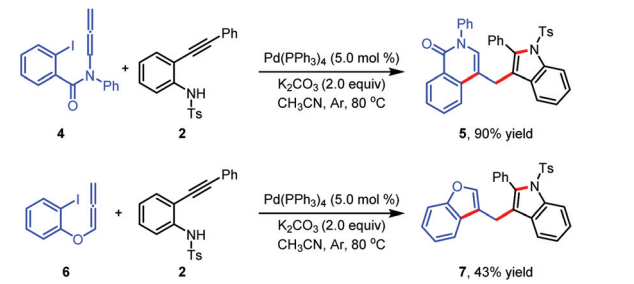
[a] Reaction conditions: **1** (0.1 mmol), **2** (0.11 mmol), $Pd(PPh_3)_4$ (5.0 mol%), K_2CO_3 (2.0 equiv.), CH_3CN (2.0 mL), at 80 °C under Ar atmosphere for 24 h, sealed tube. [b] Isolated yields.

Table 3 Scale-up experiments and synthetic transformations



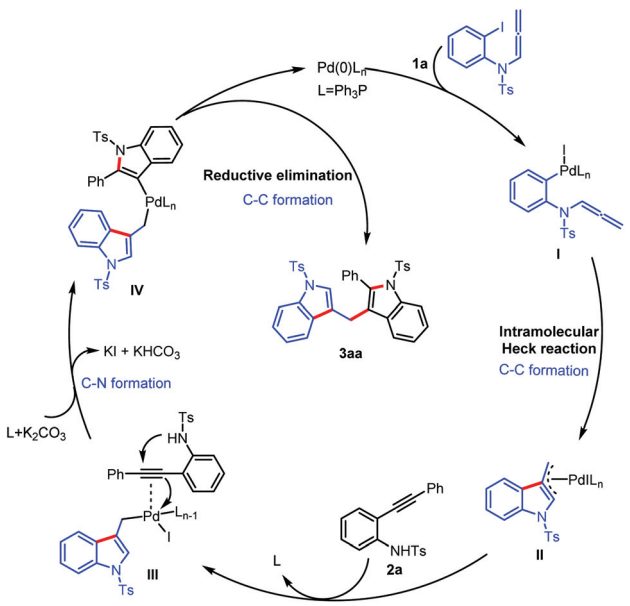
Reaction conditions: (a) R = Ph, $Pd(PPh_3)_4$ (5.0 mol%), K_2CO_3 (2.0 equiv.), CH_3CN, at 80 °C under Ar for 48 h; R = TMS, $Pd(PPh_3)_4$ (5.0 mol%), K_2CO_3 (2.0 equiv.), CH_3CN, at 80 °C under Ar for 24 h. (b) NaOH (6.0 equiv.), EtOH, at 80 °C for 18 h. (c) SeO_2 (20.0 equiv.), 1,4-dioxane, at 100 °C for 72 h. (d) $NaBH_4$ (2.0 equiv.), MeOH, 0 °C to rt, 16 h. (e) $InBr_3$ (20.0 mol%), $TMSN_3$ (3.0 equiv.), DCM, rt, 10 h. (f) $InBr_3$ (20.0 mol%), TMSCN (3.0 equiv.), DCM, rt, 10 h. (g) $InCl_3$ (20.0 mol%), TMSBr (20.0 mol%), hexane/DCM (3 : 2), at 50 °C under Ar for 2 h. (h) $InCl_3$ (20.0 mol%), TMSBr (0.6 equiv.), hexane/DCM (3 : 2), at 50 °C under Ar for 4 h.

further diversification. Subsequently, under the catalysis of Lewis acid, azido,[20] cyano,[21] allyl and alkynyl[22] functional groups could be facilely introduced into the methylene unit of DIMs *via* direct coupling reaction between alcohol **10** and silyl compounds (Table 3e–3h).



Scheme 2 Extension to the isoquinolone and benzofuran scaffolds.

On the basis of the previous literature precedents,[16,17] the plausible catalytic cycle of this reaction was depicted in Scheme 3, using allenamide **1a** and *o*-ethynylaniline **2a** exem-



Scheme 3 Proposed reaction mechanism.

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plify. Oxidative addition of allenamide **1a** to the active Pd(0) catalyst generates the Pd(II) species **I**, which undergoes an intramolecular Heck reaction to achieve the π-allylpalladium species **II**.[16] The triple bond of *o*-ethynylaniline **2a** coordinates to intermediate **II** to produce intermediate **III** by ligand exchange. In the presence of base, species **III** could easily convert to species **IV**.[17] Finally, the C–C reductive elimination of species **IV** produces diindolylmethane **3aa** and regenerates the Pd(0) species to the next catalyzed cycle.

Conclusions

In summary, we have developed a Pd-catalyzed cascade Heck/cyclization reaction of allenamides and *o*-ethynylanilines, providing a straightforward platform to prepare diversified unsymmetrically substituted 3,3′-diindolylmethanes with a modifiable methylene in high efficiency. This protocol showcases good functional group tolerance and scalability. Further transformation of the methylene unit of diindolylmethane was achieved by treatment with silyl nucleophiles and oxidant. The reaction proceeded through a sequential alkene insertion, aminopalladation and reductive elimination, involving one C–N bond, two C–C bonds and two new rings formations.

Conflicts of interest

There are no conflicts to declare.

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Notes and references

1 For selected examples, see: (*a*) Z. Huang, L. Zuo, Z. Zhang, J. Liu, J. Chen, L. Dong and J. Zhang, 3,3′-Diindolylmethane Decreases VCAM-1 Expression and Alleviates Experimental Colitis via a BRCA1-Dependent Antioxidant Pathway, *Free Radicals Biol. Med.*, 2011, **50**, 228; (*b*) E.-J. Jeon, M. Davaatseren, J.-T. Hwang, J. H. Park, H. J. Hur, A. S. Lee and M. J. Sung, Effect of Oral Administration of 3,3′-Diindolylmethane on Dextran Sodium Sulfate-Induced Acute Colitis in Mice, *J. Agric. Food Chem.*, 2016, **64**, 7702; (*c*) W.-S. Li, C.-H. Wang, S. Ko, T. T. Chang, Y. C. Jen, C.-F. Yao, S. V. More and S.-C. Jao, Synthesis and Evaluation of the Cytotoxicities of Tetraindoles: Observation that the 5-Hydroxy Tetraindole (SK228) Induces G2 Arrest and Apoptosis in Human Breast Cancer Cells, *J. Med. Chem.*, 2012, **55**, 1583; (*d*) K. V. Sashidhara, K. B. Rao, R. Sonkar, R. K. Modukuri, Y. S. Chhonker, P. Kushwaha, H. Chandasana, A. K. Khanna, R. S. Bhatta, G. Bhatia, M. K. Suthar, J. K. Saxena, V. Kumard and M. I. Siddiqi, Hybrids of Coumarin–Indole: Design, Synthesis and Biological Evaluation in Triton WR-1339 and Highfat Diet Induced Hyperlipidemic Rat Models, *MedChemComm*, 2016, **7**, 1858; (*e*) R. Clark, J. Lee and S.-H. Lee, Synergistic Anticancer Activity of Capsaicin and 3,3′-Diindolylmethane in Human Colorectal Cancer, *J. Agric. Food Chem.*, 2015, **63**, 4297; (*f*) T. Pillaiyar, M. Köse, K. Sylvester, H. Weighardt, D. Thimm, G. Borges, I. Förster, I. Kügelgen and C. E. Müller, Diindolylmethane Derivatives: Potent Agonists of the Immunostimulatory Orphan G Protein-Coupled Receptor GPR84, *J. Med. Chem.*, 2017, **60**, 3636; (*g*) M. G. Bell, K. Gavardinas, D. L. Gernert, T. A. Grese, P. K. Jadhav, P. A. Lander and M. I. Steinberg, Indole-Derivative Modulators of Steroid Hormone Nuclear Receptors, *U.S. Patent Appl* 20060235222A1 20061019, 2006.

2 For selected examples, see: (*a*) L. Wang, X. He, Y. Guo, J. Xu and S. Shao, Tris(indolyl)methene Molecule as an Anion Receptor and Colorimetric Chemosensor: Tunable Selectivity and Sensitivity for Anions, *Org. Biomol. Chem.*, 2011, **9**, 752; (*b*) A. Khorshidi, N. Mardazad and Z. Shaabanzadeh, Zirconium(IV)-Catalyzed One-Pot Synthesis and Oxidation of Bis- and Tris(indolyl)methanes into Conjugated Chromophores as New pH Indicators or Calorimetric Chemosensors for Transition Metals, *Tetrahedron Lett.*, 2014, **55**, 3873.

3 For selected examples, see: (*a*) X. He, S. Hu, K. Liu, Y. Guo, J. Xu and S. Shao, Oxidized Bis(indolyl)methane: A Simple and Efficient Chromogenic-Sensing Molecule Based on the Proton Transfer Signaling Mode, *Org. Lett.*, 2006, **8**, 333; (*b*) R. Martínez, A. Espinosa, A. Tárraga and P. Molina, Bis(indolyl)methane Derivatives as Highly Selective Colourimetric and Ratiometric Fluorescent Molecular Chemosensors for Cu^{2+} Cations, *Tetrahedron*, 2008, **64**, 2184.

4 For reviews, see: D. Sarmah and U. Bora, Methylene Surrogates for the Synthesis of 3,3′-Diindolylmethanes, *ChemistrySelect*, 2020, **5**, 8577 For selected examples of the synthesis of symmetrical DIMs via transition metal catalysis from *o*-alkynylanilines and carbonyl sources, see: (*a*) C. Praveen, Y. Wilson Sagayaraj and P. T. Perumal, Gold (I) Catalyzed Sequential Cycloisomerization/Bis-Addition of *o*-Ethynylanilines: an Efficient Access to Bis(indolyl) methanes and Di(indolyl)indolin-2-ones, *Tetrahedron Lett.*, 2009, **50**, 644; (*b*) S. Karmakar, P. Das, D. Ray, S. Ghosh and S. K. Chattopadhyay, Ag(I)-Catalyzed Domino Cyclization-Addition Sequence with Simultaneous Carbonyl and Alkyne Activation as a Route to 2,2′-Disubstituted Bisindolylarylmethanes, *Org. Lett.*, 2016, **18**, 5200.

5 For selected examples, see: (*a*) A. K. Chakraborti, S. R. Roy, D. Kumar and P. Chopra, Catalytic Application of Room

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Temperature Ionic Liquids: [bmim][$MeSO_4$] as a Recyclable Catalyst for Synthesis of Bis(indolyl)methanes. Ion-Fishing by MALDI-TOF-TOF MS and MS/MS Studies to Probe the Proposed Mechanistic Model of Catalysis, *Green Chem.*, 2008, **10**, 1111; (*b*) T. Abe, S. Nakamura, R. Yanada, T. Choshi, S. Hibino and M. Ishikura, One-Pot Construction of 3,3′-Bisindolylmethanes through Bartoli Indole Synthesis, *Org. Lett.*, 2013, **15**, 3622; (*c*) K. Griffiths, P. Kumar, G. R. Akien, N. F. Chilton, A. Abdul-Sada, G. J. Tizzard, S. J. Coles and G. E. Kostakis, Tetranuclear Zn/4f Coordination Clusters as Highly Efficient Catalysts for Friedel-Crafts Alkylation, *Chem. Commun.*, 2016, **52**, 7866; (*d*) F. Lafzi, H. Kilic and N. Saracoglu, Protocols for the Syntheses of 2,2′-Bis(indolyl)arylmethanes, 2-Benzylated Indoles, and 5,7-Dihydroindolo[2,3-*b*]carbazoles, *J. Org. Chem.*, 2019, **84**, 12120; (*e*) T. Yang, H. Lu, Y. Shu, Y. Ou, L. Hong, C.-T. Au and R. Qiu, CF_3SO_2Na-Mediated, UV-Light-Induced Friedel-Crafts Alkylation of Indoles with Ketones/Aldehydes and Bioactivities of Products, *Org. Lett.*, 2020, **22**, 827.

6 J. Xiang, J. Wang, M. Wang, X. Meng and A. Wu, One-pot Total Synthesis of Streptindole, Arsindoline B and Their Congeners through Tandem Decarboxylative Deaminative Dual-Coupling Reaction of Amino Acids with Indoles, *Org. Biomol. Chem.*, 2015, **13**, 4240.

7 (*a*) H. Yu and Z. Yu, Direct Alkenylation of Indoles with α-Oxo Ketene Dithioacetals: Efficient Synthesis of Indole Alkaloids Meridianin Derivatives, *Angew. Chem., Int. Ed.*, 2009, **48**, 2929; (*b*) S. Lucarini, M. Mari, G. Piersanti and G. Spadoni, Organocatalyzed Coupling of Indoles with Dehydroalanine Esters: Synthesis of Bis(indolyl)propanoates and Indolacrylates, *RSC Adv.*, 2013, **3**, 19135; (*c*) H.-Y. Xu, Y. Zi, X.-P. Xu, S.-Y. Wang and S.-J. Ji, TFA-Catalyzed C-N Bond Activation of Enamides with Indoles: Efficient Synthesis of 3,3-Bisindolylpropanoates and Other Bisindolylalkanes, *Tetrahedron*, 2013, **69**, 1600.

8 S. Ma and S. Yu, Sc$(OTf)_3$-C atalyzed Indolylation of 1,2-Allenic Ketones: Controlled Highly Selective Synthesis of *β*-Indolyl-*α*,*β*-Unsaturetated (*E*)-Enones and *β*,*β*-Bisindolyl Ketones, *Org. Lett.*, 2005, **7**, 5063.

9 For selected examples, see: (*a*) M.-H. Xie, F.-D. Xie, G.-F. Lin and J.-H. Zhang, Convenient Synthesis of Bis (indolyl)alkanes and Bis(pyrrolyl)alkanes by Cu$(OTf)_2$-C atalyzed Addition of Indole and Pyrrole to Acetylenic Sulfone, *Tetrahedron Lett.*, 2010, **51**, 1213; (*b*) T. Tsuchimoto and M. Kanbara, Reductive Alkylation of Indoles with Alkynes and Hydrosilanes under Indium Catalysis, *Org. Lett.*, 2011, **13**, 912.

10 For selected examples, see: (*a*) S. D. Jadhav, D. Bakshi and A. Singh, Visible Light Mediated Organocatalytic Activation of Ethyl Bromofluoroacetate: Coupling with Indoles and Anilines, *J. Org. Chem.*, 2015, **80**, 10187; (*b*) L. Zheng, F. Gao, C. Yang, G.-L. Gao, Y. Zhao, Y. Gao and W. Xia, Visible-Light-Mediated Anti-Regioselective Nitrone 1,3-Dipolar Cycloaddition Reaction and Synthesis of Bisindolylmethanes, *Org. Lett.*, 2017, **19**, 5086; (*c*) L. Ye, S.-H. Cai, D.-X. Wang, Y.-Q. Wang, L.-J. Lai, C. Feng and T.-P. Loh, Photoredox Catalysis Induced Bisindolylation of Ethers/Alcohols via Sequential C-H and C-O Bond Cleavage, *Org. Lett.*, 2017, **19**, 6164; (*d*) Y. Zhang, X. Yang, H. Zhou, S. Li, Y. Zhu and Y. Li, Visible Light-Induced Aerobic Oxidative Cross-Coupling of Glycine Derivatives with Indoles: a Facile Access to 3,3′ Bisindolylmethanes, *Org. Chem. Front.*, 2018, **5**, 2120.

11 For selected examples, see: (*a*) X. Guo, S. Pan, J. Liu and Z. Li, One-Pot Synthesis of Symmetric and Unsymmetric 1,1-Bis-indolylmethanes via Tandem Iron-Catalyzed C-H Bond Oxidation and C-O Bond Cleavage, *J. Org. Chem.*, 2009, **74**, 8848; (*b*) D. Xia, Y. Wang, Z. Du, Q.-Y. Zheng and C. Wang, Rhenium-Catalyzed Regiodivergent Addition of Indoles to Terminal Alkynes, *Org. Lett.*, 2012, **14**, 588; (*c*) P. Li, Y. Weng, X. Xu and X. Cui, Access to Indole Derivatives from Diaryliodonium Salts and 2-Alkynylanilines, *J. Org. Chem.*, 2016, **81**, 3994; (*d*) A. Srivastava, S. S. Patel, N. Chandna and N. Jain, Copper-Catalyzed *anti*-Markovnikov Hydroindolation of Terminal Alkynes: Regioselective Synthesis of Bis(indolyl) alkanes, *J. Org. Chem.*, 2016, **81**, 11664; (*e*) S. Zhang, Z. Chen, S. Qin, C. Lou, A. M. Senan, R.-Z. Liao and G. Yin, Non-Redox Metal Ion Promoted Oxidative Coupling of Indoles with Olefins by the Palladium(II) Acetate Catalyst through Dioxygen Activation: Experimental Results with DFT Calculations, *Org. Biomol. Chem.*, 2016, **14**, 4146.

12 Z.-B. Xie, D.-Z. Sun, G.-F. Jiang and Z.-G. Le, Facile Synthesis of Bis(indolyl)methanes Catalyzed by α-Chymotrypsin, *Molecules*, 2014, **19**, 19665.

13 K.-S. Du and J.-M. Huang, Electrochemical Synthesis of Bisindolylmethanes from Indoles and Ethers, *Org. Lett.*, 2018, **20**, 2911.

14 For selected examples, see: (*a*) H. M. Kaiser, W. F. Lo, A. M. Riahi, A. Spannenberg, M. Beller and M. K. Tse, New Synthetic Protocols for the Preparation of Unsymmetrical Bisindoles, *Org. Lett.*, 2006, **8**, 5761; (*b*) G. Herrán, A. Segura and A. G. Csákÿ, Benzylic Substitution of Gramines with Boronic Acids and Rhodium or Iridium Catalysts, *Org. Lett.*, 2007, **9**, 961; (*c*) M.-H. Zhuo, Y.-J. Jiang, Y.-S. Fan, Y. Gao, S. Liu and S. Zhang, Enantioselective Synthesis of Triarylmethanes by Chiral Imidodiphosphoric Acids Catalyzed Friedel-Crafts Reactions, *Org. Lett.*, 2014, **16**, 1096; (*d*) L. P. Lin, P. Yuan, N. Jiang, Y. N. Mei, W. J. Zhang, H. M. Wu, A. H. Zhang, J. M. Cao, Z. X. Xiong, Y. Lu and R. X. Tan, Gene-Inspired Mycosynthesis of Skeletally New Indole Alkaloids, *Org. Lett.*, 2015, **17**, 2610; (*e*) J. Xiao, H. Wen, L. Wang, L. Xu, Z. Hao, C.-L. Shao and C.-Y. Wang, Catalyst-Free Dehydrative S_N1-Type Reaction of Indolyl Alcohols with Diverse Nucleophiles "on Water", *Green Chem.*, 2016, **18**, 1032; (*f*) Y. Ling, D. An, Y. Zhou and W. Rao, Ga$(OTf)_3$-C atalyzed Temperature-Controlled Regioselective Friedel-Crafts Alkylation of Trifluoromethylated 3-Indolylmethanols with 2-Substituted Indoles: Divergent Synthesis of Trifluoromethylated

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Unsymmetrical 3,3′-and 3,6′-Bis(indolyl)methanes, *Org. Lett.*, 2019, **21**, 3396.

15 (*a*) S. Guo, K. Yuan, M. Gu, A. Lin and H. Yao, Rh(III)-Catalyzed Cascade Annulation/C-H Activation of *o*-Ethynylanilines with Diazo Compounds: One-Pot Synthesis of Benzo[*a*]carbazoles via 1,4-Rhodium Migration, *Org. Lett.*, 2016, **18**, 5236; (*b*) S. Guo, R. Pan, Z. Guan, P. Li, L. Cai, S. Chen, A. Lin and H. Yao, Synthesis of Indole-Fused Polycyclics via Rhodium-Catalyzed Undirected C-H Activation/Alkene Insertion, *Org. Lett.*, 2019, **21**, 6320; (*c*) S. Guo, P. Li, Z. Guan, L. Cai, S. Chen, A. Lin and H. Yao, Pd-Catalyzed Alkyne Insertion/C-H Activation/[4 + 2] Carboannulation of Alkenes to the Synthesis of Polycyclics, *Org. Lett.*, 2019, **21**, 921.

16 For selected examples of allenamides were used as indole precursors to generate in situ indole skeletons, see: (*a*) S. Husinec, M. Petkovic, V. Savic and M. Simic, Synthesis of Allyl Acetates via Palladium-Catalysed Functionalisation of Allenes and 1,3-Dienes, *Synthesis*, 2012, **44**, 399; (*b*) M.-G. Braun, M. H. Katcher and A. G. Doyle, Carbofluorination via a Palladium-Catalyzed Cascade Reaction, *Chem. Sci.*, 2013, **4**, 1216; (*c*) Y. Higuchi, T. Mita and Y. Sato, Palladium-Catalyzed Intramolecular Arylative Carboxylation of Allenes with CO_2 for the Construction of 3-Substituted Indole-2-Carboxylic Acids, *Org. Lett.*, 2017, **19**, 2710; (*d*) J. Hédouin, C. Schneider, I. Gillaizeau and C. Hoarau, Palladium-Catalyzed Domino Allenamide Carbopalladation/Direct C-H Allylation of Heteroarenes: Synthesis of Primprinine and Papaverine Analogues, *Org. Lett.*, 2018, **20**, 6027; (*e*) X. Chen, G. Qiu, R. Liu, D. Chen and Z. Chen, Divergent Oriented Synthesis (DOS) of Azaheterocyclic Amides Through Palladium-Catalyzed Ketenimination of 2-Iodo-*N*-(propa-1,2-dien-1-yl) Anilines, *Org. Chem. Front.*, 2020, **7**, 890; (*f*) W. Xiong, R. Cheng, B. Wu, W. Wu, C. Qi and H. Jiang, Palladium-Catalyzed Regioselective Cascade Reaction of Carbon Dioxide, Amines and Allenes for the Synthesis of Functionalized Carbamates, *Sci. China: Chem.*, 2020, **63**, 331.

17 For selected examples of *o*-ethynylanilines were used as indole precursors to generate in situ indole skeletons, see: (*a*) C.-C. Chen, L.-Y. Chin, S.-C. Yang and M.-J. Wu, Synthetic Development and Mechanistic Study on Pd(II)-Catalyzed Cyclization of Enediynes to Benzo[*a*]carbazoles, *Org. Lett.*, 2010, **12**, 5652; (*b*) X.-F. Xia, N. Wang, L.-L. Zhang, X.-R. Song, X.-Y. Liu and Y.-M. Liang, Palladium(II)-Catalyzed Tandem Cyclization/C-H Functionalization of Alkynes for the Synthesis of Functionalized Indoles, *J. Org. Chem.*, 2012, **77**, 9163; (*c*) M. Tian, D. Bai, G. Zheng, J. Chang and X. Li, Rh(III)-Catalyzed Asymmetric Synthesis of Axially Chiral Biindolyls by Merging C-H Activation and Nucleophilic Cyclization, *J. Am. Chem. Soc.*, 2019, **141**, 9527.

18 S. E. Dibrell, M. R. Maser and S. E. Reisman, SeO_2-Mediated Oxidative Transposition of Pauson-Khand Products, *J. Am. Chem. Soc.*, 2020, **142**, 6483.

19 T. W. Bevan, J. Francis-Taylor, H. Wong, P. T. Northcote and J. E. Harvey, A Colourful Azulene-Based Protecting Group for Carboxylic Acids, *Tetrahedron*, 2018, **74**, 2942.

20 A. Kumar, R. K. Sharma, T. V. Singh and P. Venugopalan, Indium(III) Bromide Catalyzed Direct Azidation of α-Hydroxyketones Using $TMSN_3$, *Tetrahedron*, 2013, **69**, 10724.

21 G. Chen, Z. Wang, J. Wu and K. Ding, Facile Preparation of α-Aryl Nitriles by Direct Cyanation of Alcohols with TMSCN Under the Catalysis of InX_3, *Org. Lett.*, 2008, **10**, 4573.

22 T. Saito, Y. Nishimoto, M. Yasuda and A. Baba, Direct Coupling Reaction between Alcohols and Silyl Compounds: Enhancement of Lewis Acidity of Me_3SiBr Using $InCl_3$, *J. Org. Chem.*, 2006, **71**, 8516.

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